Electrotherapy

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Course Description (Electrotherapy) (2019 – 2020)

1. Electromagnetic spectrum
2. **organization of body system**
3. **pain**
4. **Infrared+ Physiologic effect of heat (Quiz 1)**
5. Short wave

Course Description (Electrotherapy) (2019 – 2020)

6	SW + microwave *(Quiz 2)*
7	Ultrasound
8	Phonophoresis
9	ultraviolet (Quiz 3)
10 11	LASER , Proplem solving

Course Description (Electrotherapy) (2019 – 2020)

12 Shock wave

13 Magnetic therapy

14 Biofeedback

15 Proplem solving

Electromagnetic spectrum

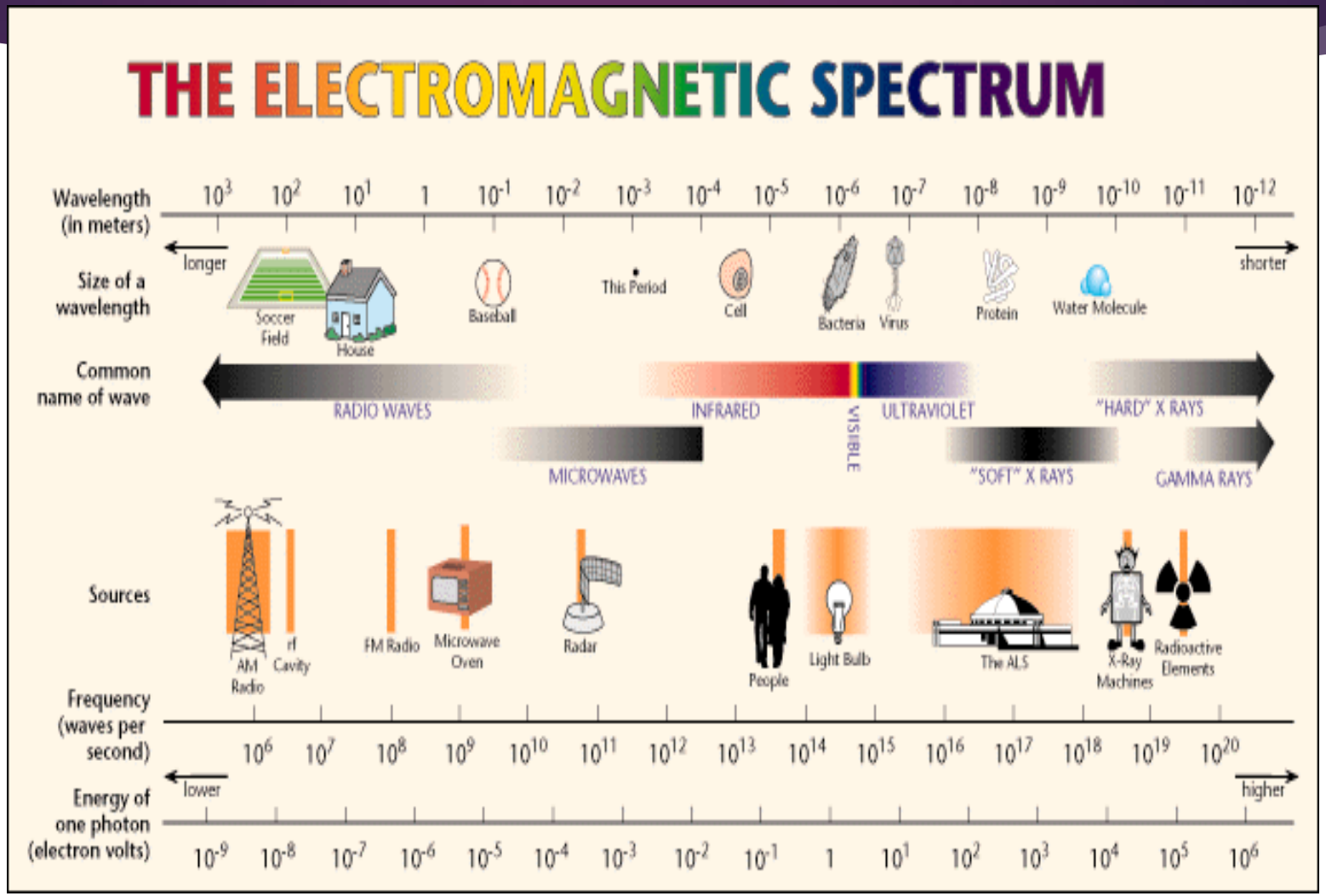
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Electromagnetic spectrum

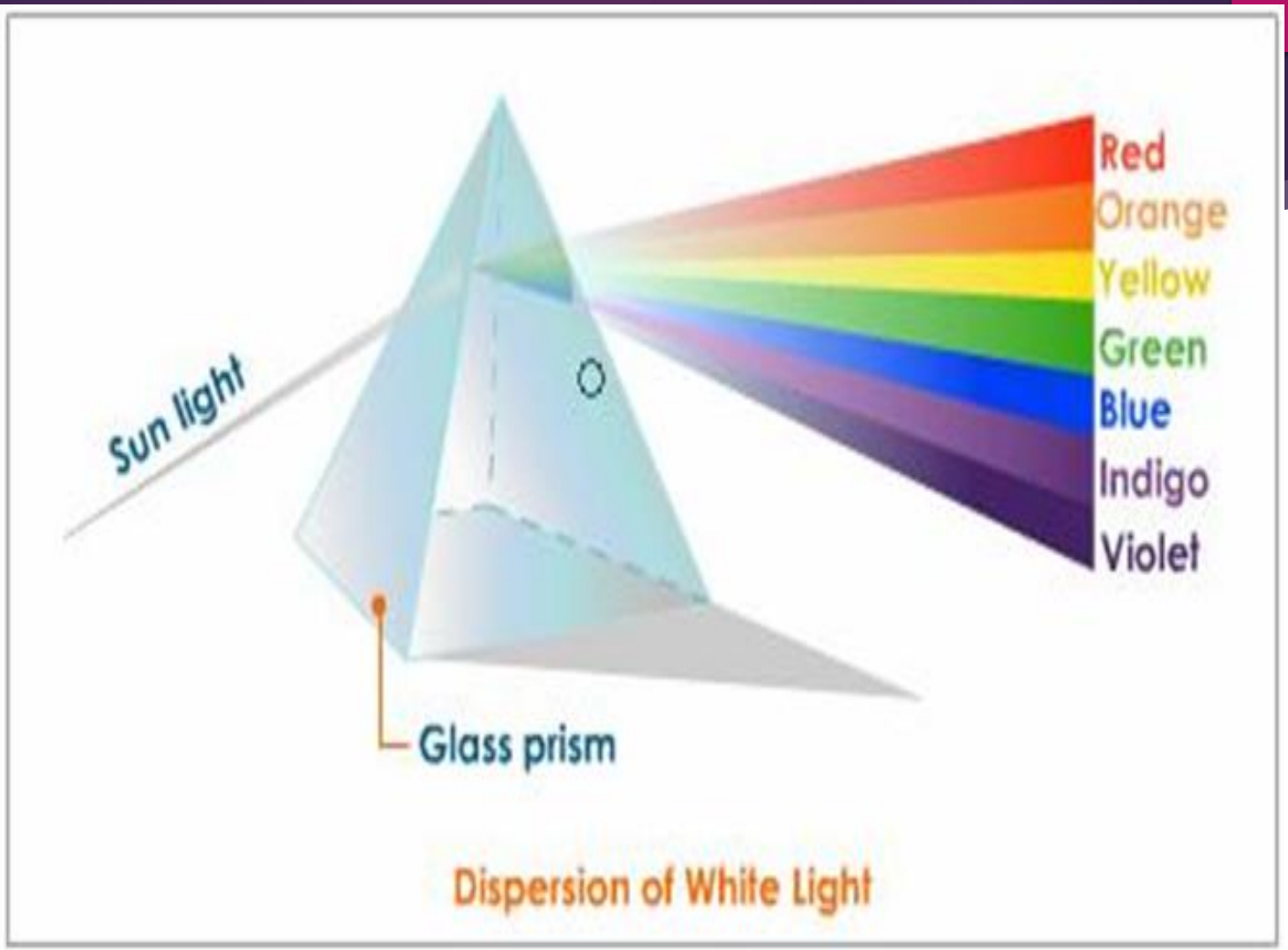
- **Electromagnetic waves** are waves that are capable of traveling through a vacuum. Unlike **mechanical waves** that require a medium in order to transport their energy, electromagnetic waves are capable of transporting energy through the vacuum of outer space.

Electromagnetic spectrum



The Electromagnetic Energy Spectrum

- **If a ray of sunlight is passed through a prism, it will be broken down into various colors in a predictable rainbow-like pattern of red, orange, yellow, green, blue, indigo, and violet. The range of colors is called a spectrum.**

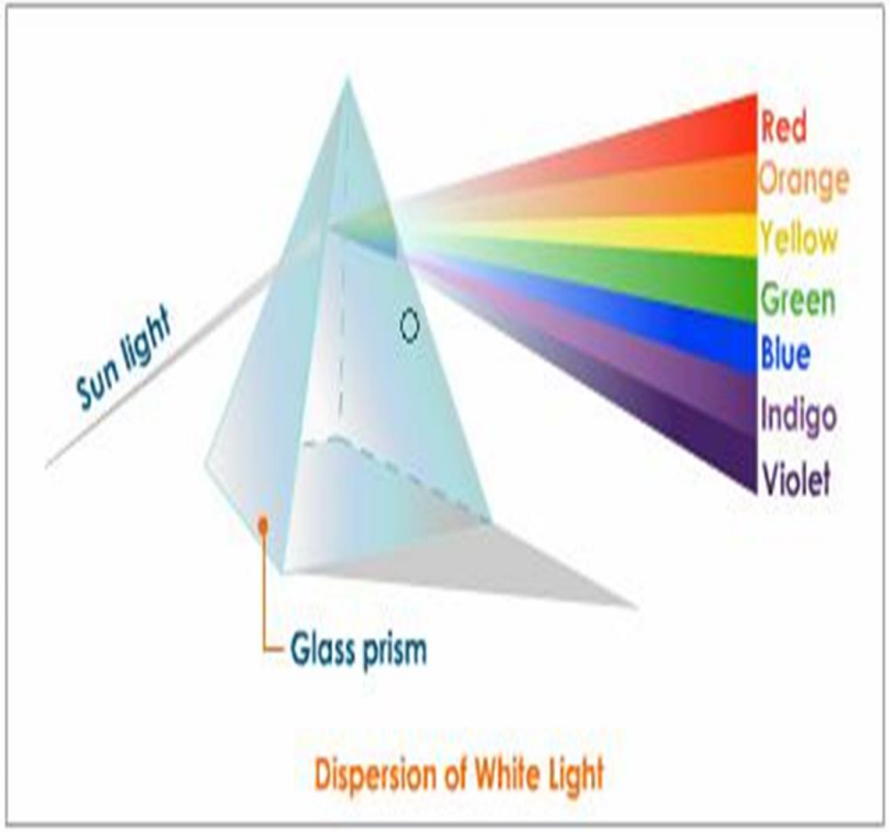

Red
Orange
Yellow
Green
Blue
Indigo
Violet
Sun light
Glass prism
Dispersion of White Light

The Electromagnetic Spectrum Energy

- The colors that we can detect with our eyes are referred to as *visible light* or *luminous radiations*. Each of these colors represents a photon of a different energy. They appear as different colors because the various forms of radiant energy are refracted or change direction as a result of differences in wavelength and frequency of each color.

The Electromagnetic Energy Spectrum

- **The longest wavelength light is red in color and low in energy, whereas the shortest wavelength light is violet and relatively higher in energy.**



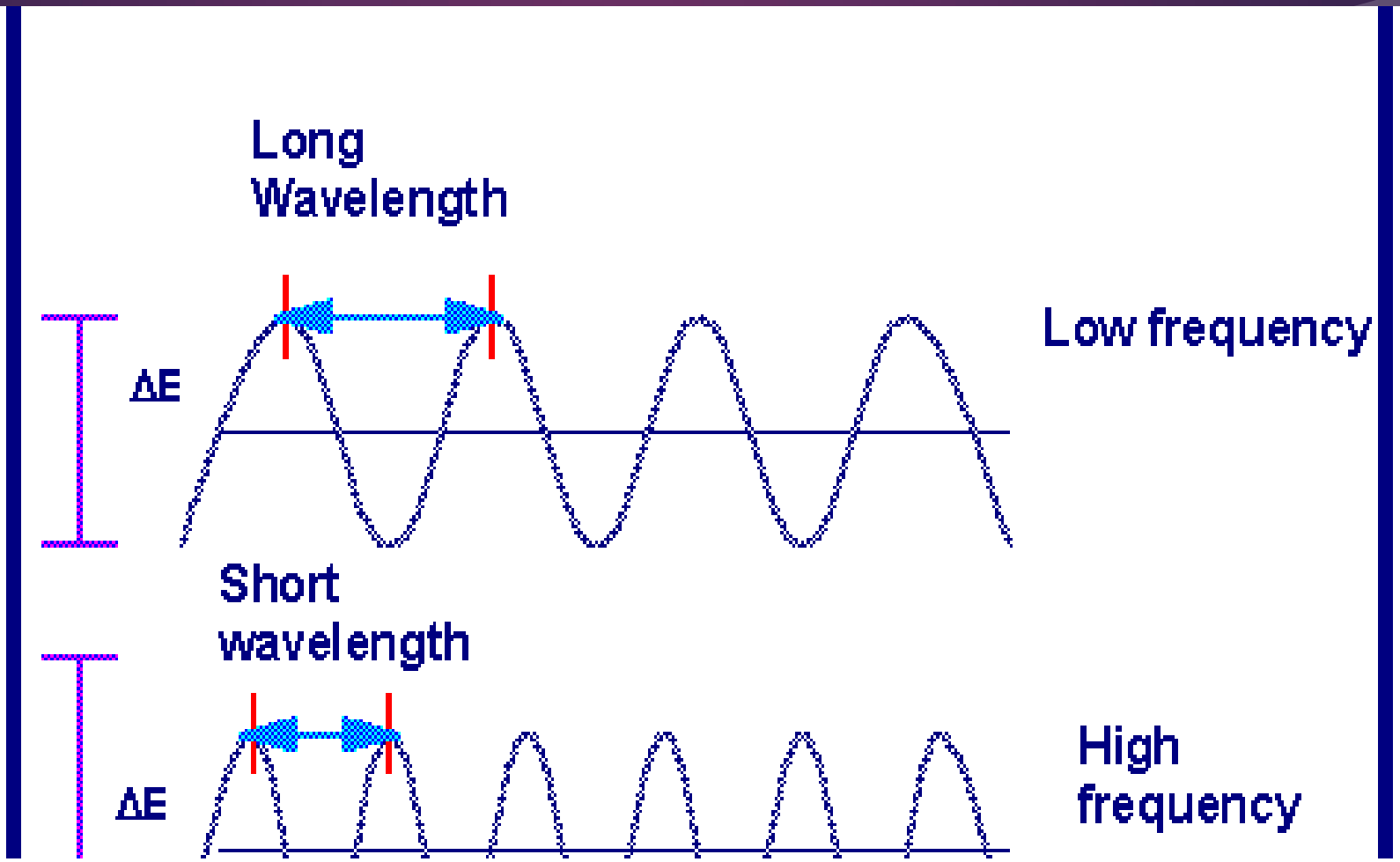
Effects of Electromagnetic Radiations

- The effects of electromagnetic radiation on tissues depend on the wavelength, frequency, and energy of the electromagnetic waves that penetrate those tissues.
- The electromagnetic radiation with higher amounts of energy per photon can have different, more dramatic effects on tissues.(diathermies)
- Ultraviolet radiation is more energetic than visible light and carries enough energy to damage tissues.

The Relationship between Wavelength and Frequency

- **Wavelength** is defined as the distance between the peak of one wave and the peak of either the preceding or succeeding wave.
- **Frequency** is defined as the number of wave oscillations or vibrations occurring in a particular time unit and is commonly expressed in Hertz (Hz). One Hertz is one vibration per second.

The Relationship between Wavelength and Frequency



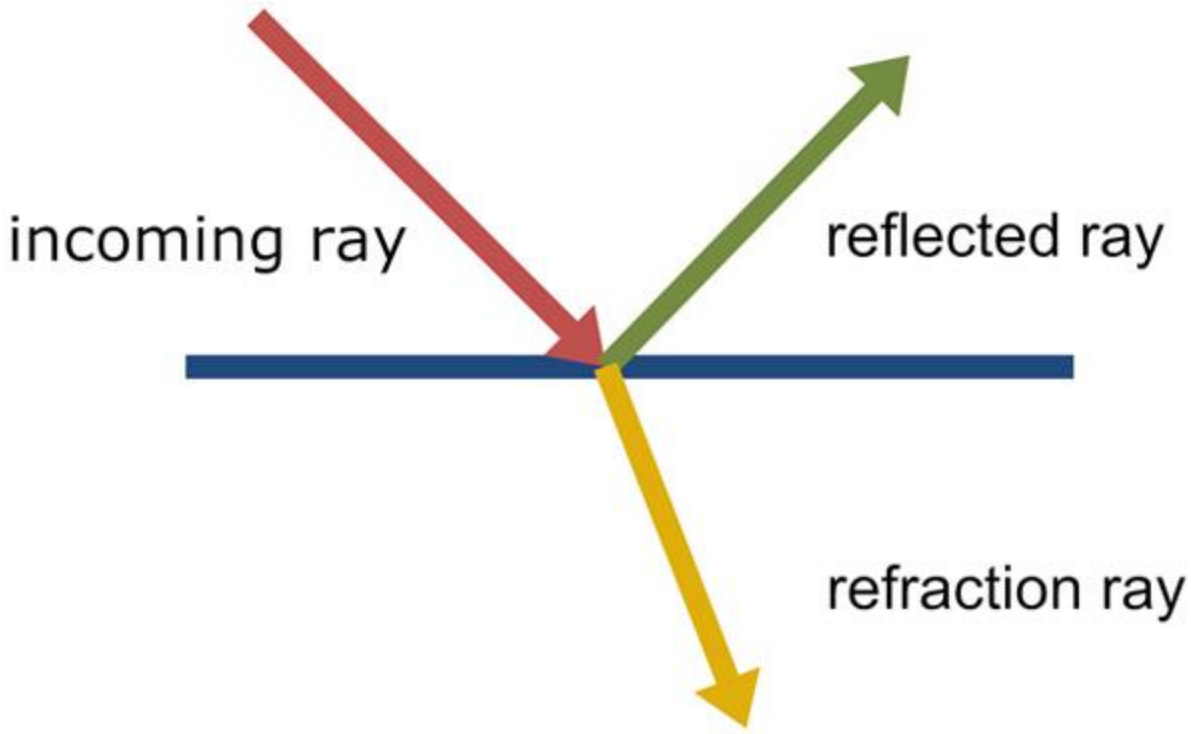
The Relationship between Wavelength and Frequency

- Since all forms of electromagnetic radiation travel at a constant velocity through space, photons with longer wavelengths have lower frequencies and photons with shorter wavelengths have higher frequencies.
- **Speed = wavelength × frequency**
- **An inverse relationship exists between wavelength and frequency. The longer the wavelength of a wave, the lower the frequency of the wave has to be.**

Laws Governing the Effects of Electromagnetic Energy

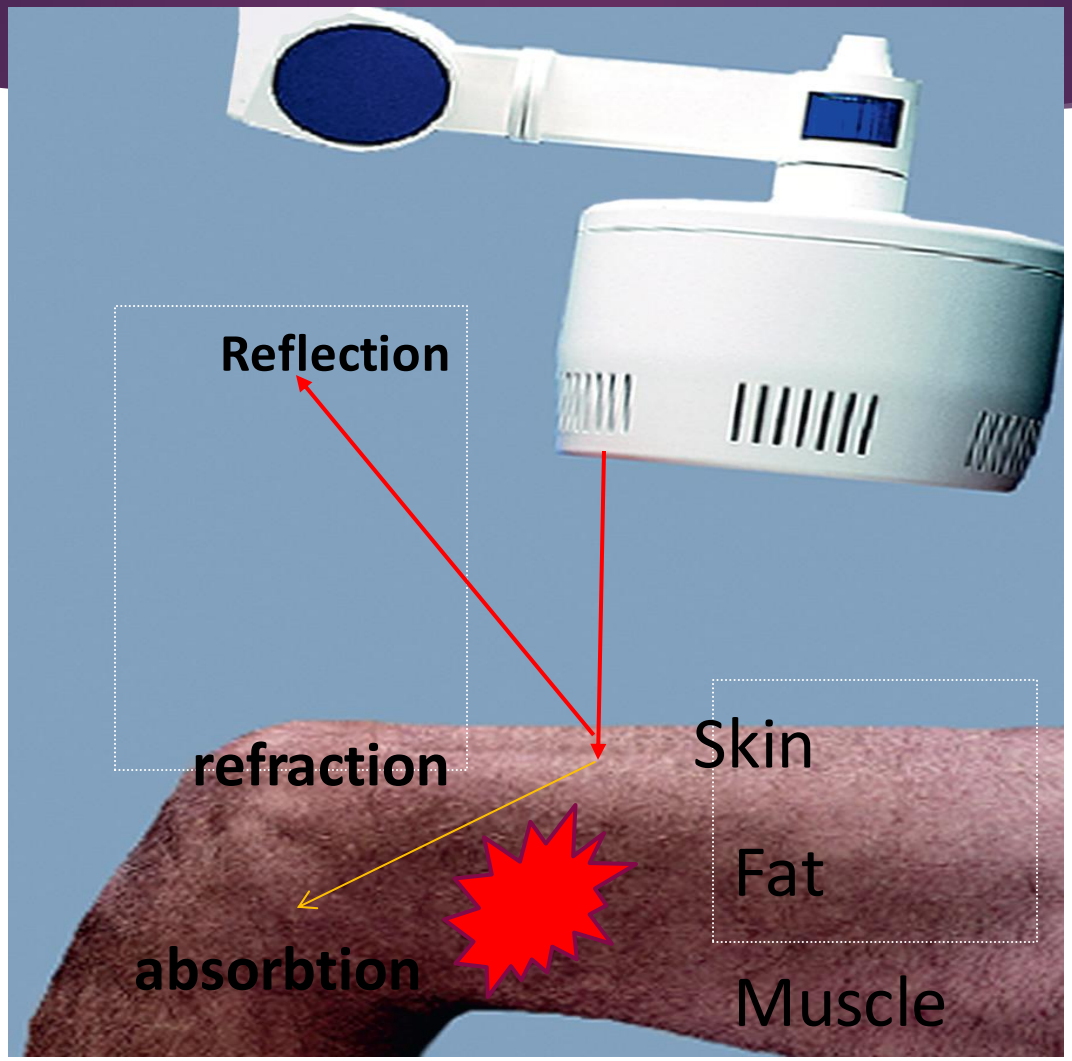
- When electromagnetic radiation strikes or comes in contact with various objects, it can be reflected, transmitted, refracted, or absorbed, depending on the type of radiation and the nature of the object it interacts with.

Reflection and refraction



Arndt–Schultz Principle

- The **Arndt–Schultz principle** states that no reactions or changes can occur in the body tissues if the amount of energy absorbed is **insufficient** to stimulate the absorbing tissues.


Reflection
refraction
absorbtion
Skin
Fat
Muscle

Law of Grotthus–Draper

- The inverse relationship that exists between energy absorption by a tissue and energy penetration to deeper layers is described by the **Law of Grotthus–Drape**
- If too much radiation is absorbed by superficial tissues, not enough will be absorbed by the deeper tissues to stimulate the tissues.

Law of Grotthus–Draper

- ultrasound at a frequency of either 1 MHz (long wavelength) or 3 MHz (short wavelength). Using ultrasound treatment at a frequency of 1 MHz would be more effective at penetrating the deeper tissues than ultrasound treatment at 3 MHz,
- since less energy would be absorbed superficially for the longer wavelength.

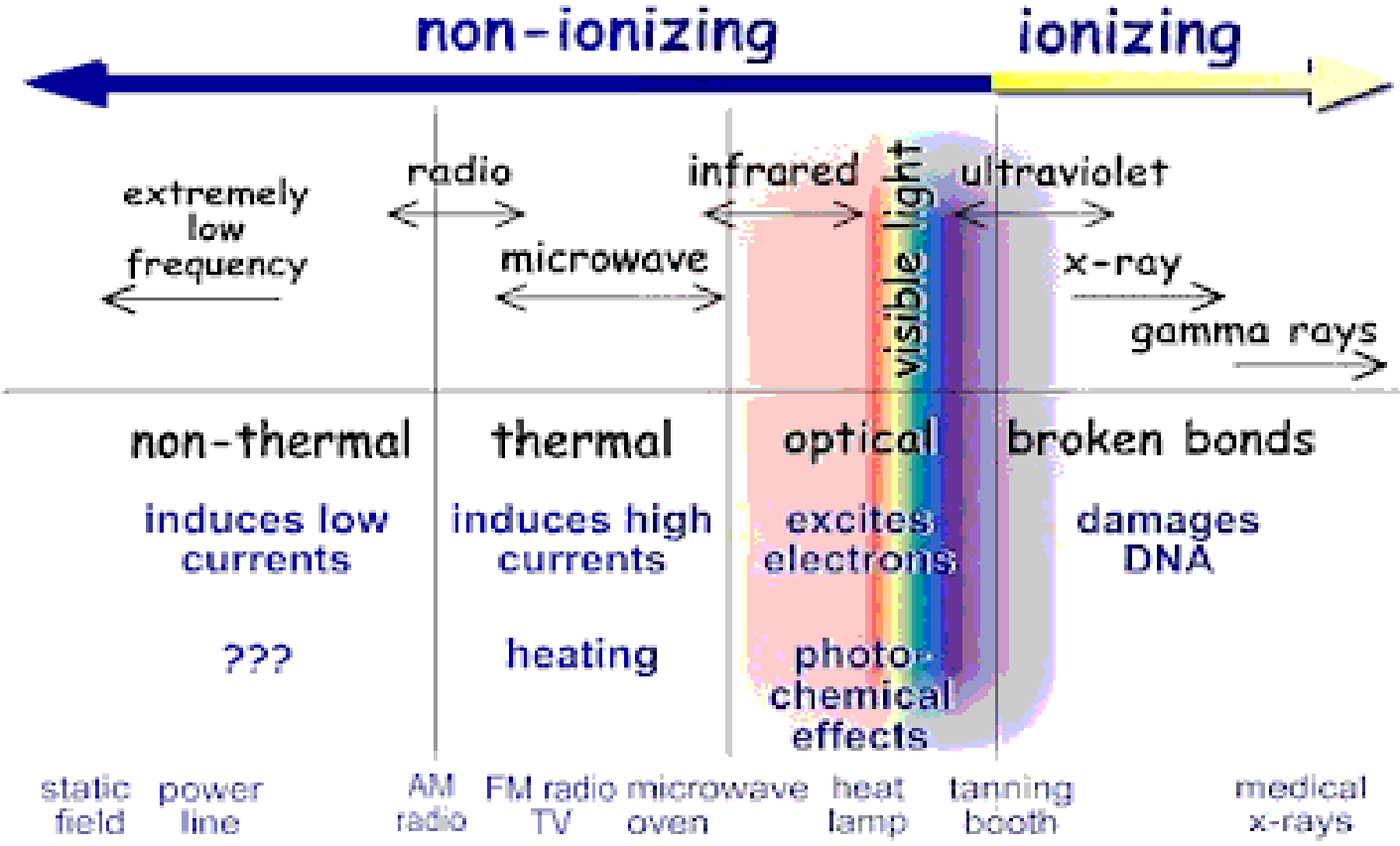
Non ionizing and ionizing range

- Near ultraviolet, visible light, infrared, microwave, radio waves, and low-frequency radio frequency (longwave) are all examples of **non-ionizing** radiation. By contrast, far ultraviolet light, X-rays, gamma-rays, and all particle radiation from radioactive decay are **ionizing**.

What is the difference between **Non-Ionizing** and **Ionizing radiation**?

- **Non-Ionizing radiation** does not carry enough energy to break molecular bonds and **ionize** atoms. . **Ionizing radiation** is the type of **radiation** that carries enough energy to break bonds **between** molecules and **ionize** atoms.

ionizing range and non ionizing range



ionizing range and non ionizing range

- **ionizing range:**
- Energy in the ionizing range can readily alter the components of atoms (electrons, protons,and neutrons). This radiation can easily penetrate tissue to deposit energy within the cells. (x ray)

- **non ionizing range:**
- Energy in the nonionizing range is commonly used in the management of musculoskeletal injuries.

Transfer of Energy

- *Radiation*
- *Conduction*
- *Convection*
- *Conversion*
- *Evaporation*

Radiation

- **Radiation** is the transfer of energy in the form of infrared waves (radiant energy) **without physical contact**. All matter radiates energy in the form of heat. Usually, body heat is warmer than the environment, and radiant heat energy is dissipated through the air to surrounding solid, cooler objects.

Radiation



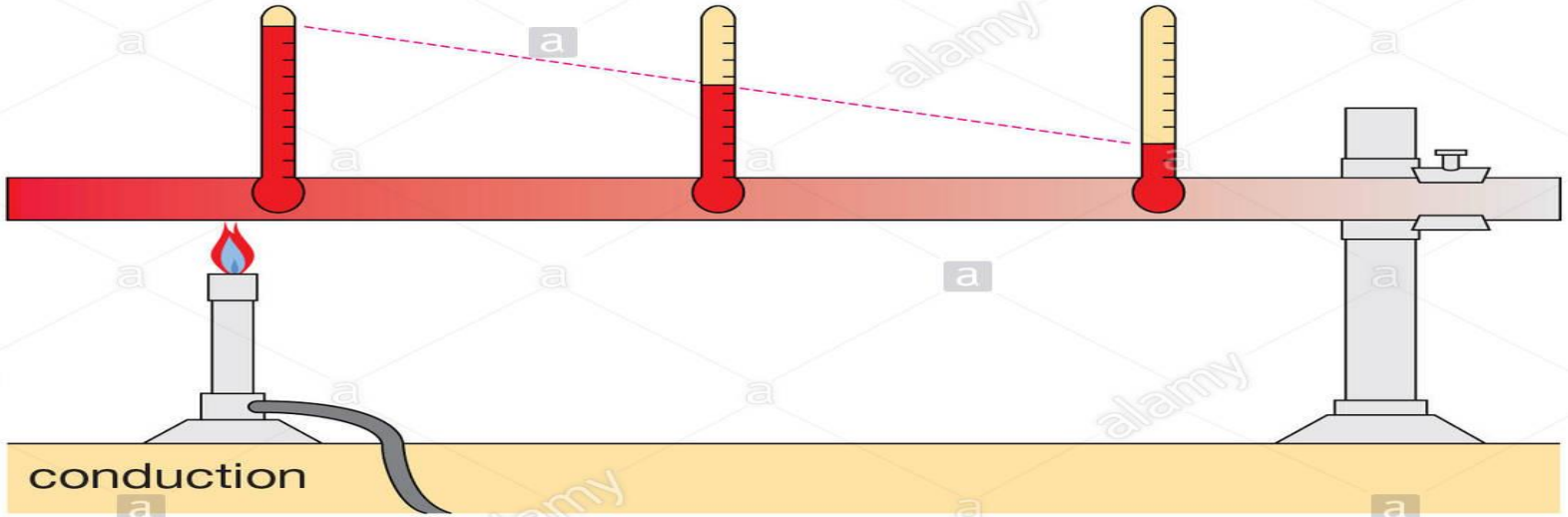
Conduction

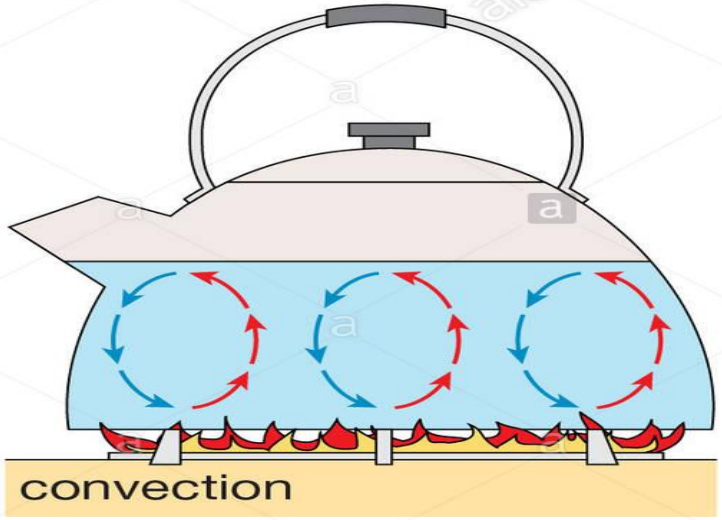
- **Conduction** is the direct transfer of energy between two objects in physical contact with each other. A difference in temperature is necessary to initiate the movement of kinetic energy from one molecule to another.

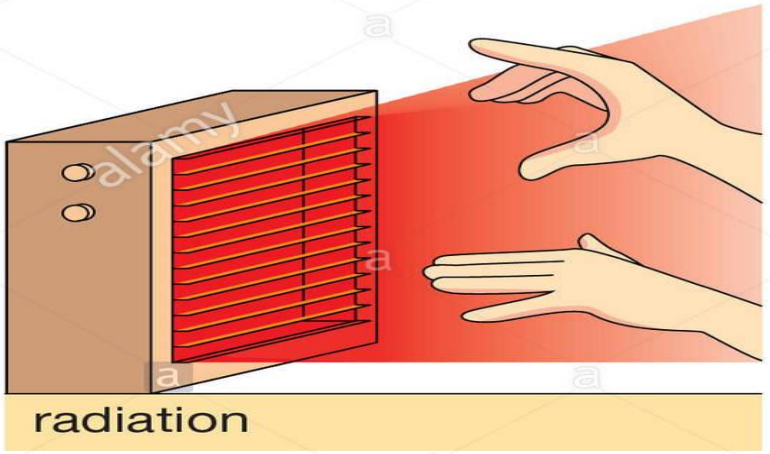
Convection

- **Convection**, a more rapid process than conduction, occurs when a medium, such as air or water, moves across the body, creating temperature variations. The effectiveness of heat loss or heat gain by conduction depends on how fast the air (or water) next to the body is moved away once it becomes warmed.

Three methods of heat transfer



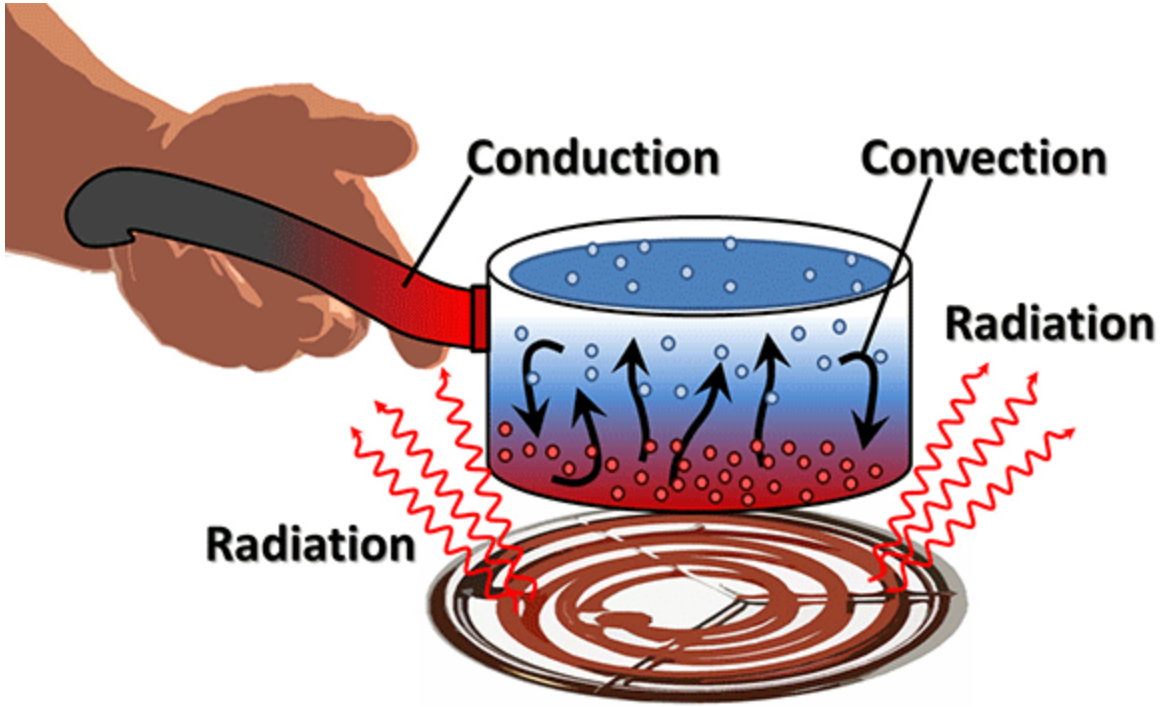




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Conversion

- **Conversion** involves the changing of another energy form (e.g., sound, electricity, or a chemical agent) into heat.
- In ultrasound therapy, mechanical energy produced by high-frequency sound waves is converted to heat energy at tissue interfaces.


Conduction
Convection
Radiation
Radiation

Evaporation



Evaporation

- Heat loss also can occur during evaporation. Vapocoolant sprays, for example, spread a liquid over the skin surface. The heat absorbed by the liquid cools the skin surface as the liquid changes into a gaseous state. Evaporation is also the means by which the body cools itself on a hot day through the evaporation of sweat.

Thank you